ANKUSH MANDAL

Computational Astrophysicist

Protoplanetary Disks • Self-Gravity • Gravitational Instability • Numerical Methods

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RESEARCH PROFILE

I am a computational astrophysicist specializing in self-gravitating astrophysical flows, protoplanetary disks, gravitational instability, and high-performance numerical methods. My current research combines global hydrodynamic and magnetohydrodynamic simulations with the development of numerical methods for self-gravity, with a particular focus on disk fragmentation, planet formation, and the thermal and magnetic regulation of gravitational instability. My broader research experience spans molecular clouds, turbulence-regulated star formation, and AGN feedback.

RESEARCH EXPERIENCE

Postdoctoral Researcher *2024–Present*

Leibniz Institute for Astrophysics Potsdam (AIP), Germany *Magnetohydrodynamics and Turbulence Section*

- Global 3D simulations of protoplanetary disks, focusing on disk self-gravity, magnetic fields, gravitational instability, fragmentation, and planet formation.
- Developed a fast hybrid spectral–multigrid Poisson solver for self-gravity in cylindrical and spherical geometries and integrated it into the Pluto code.
- Develop and optimize numerical algorithms for large-scale hydrodynamic/MHD simulations on high-performance computing systems.

Doctoral Researcher in Astrophysics *2019–2024*

Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India

- Studied the impact of AGN-driven outflows and relativistic jets on multiphase gas, molecular clouds, turbulence, and star formation.
- Developed a novel subgrid framework for estimating star formation rates in large-scale simulations, incorporating the effects of local gas turbulence and thermodynamics beyond conventional density-threshold-based prescriptions prescriptions.
- Developed numerical infrastructure for self-gravity and star-formation modelling, including Poisson solvers, sink particles for gravitationally collapsing structures, and a subgrid star-cluster formation module.

Visiting Researcher – Future Research Talent (FRT) Program *May–July 2019*

Australian National University (ANU), Canberra, Australia *Host: Dr. Christoph Federrath*

- Selected through the competitive Future Research Talent program for a three-month collaborative research project.
- Conducted computational studies of the formation of star-forming molecular clouds through large-scale compression of magnetized, turbulent interstellar gas subject to radiative cooling.

NUMERICAL METHODS & COMPUTATIONAL EXPERTISE

Self-gravity & Poisson solvers	Developed a 3D multigrid (with Runge-Kutta-Legendre acceleration) Poisson solver for Cartesian grids, and a fast hybrid spectral-multigrid solver for non-Cartesian geometries.
Pluto development	Extensive experience developing and extending Pluto, including its hydrodynamic/MHD algorithms, reconstruction, Riemann solvers, time integration, parallel decomposition, and the implementation of new physics and numerical modules.
Sink particles	Developed a sink-particle module for modelling gravitationally bound objects (stars/planets), and a star-cluster module for large-scale simulations based on a novel turbulence-regulated subgrid model of star formation.
Structure identification	Developed a parallel FellWalker-based ClumpFind algorithm for identifying and characterizing dense structures in multidimensional simulation data.

High-performance computing	Extensive experience in distributed-memory parallel programming with MPI, including the development and optimization of large-scale numerical algorithms for HPC systems.
Scientific computing	C/C++, Python, Fortran; Pluto, Flash; MPI, OpenMP, CUDA; FFTW, cuFFT, GSL; VisIt, ParaView.

PUBLICATIONS

First-author publications

A. Mandal, O. Gressel, U. Ziegler, and A. Mignone (2026), "A Fast Spectral-Multigrid Poisson Solver in Non-Cartesian Geometries," *The Astrophysical Journal Supplement Series*. [DOI].

A. Mandal, D. Mukherjee, C. Federrath, G. V. Bicknell, N. P. H. Nesvadba, and A. Mignone (2024), "Probing the role of self-gravity in clouds impacted by AGN-driven winds," *Monthly Notices of the Royal Astronomical Society*. [DOI].

A. Mandal, D. Mukherjee, and A. Mignone (2023), "A self-gravity module for the PLUTO code," *The Astrophysical Journal Supplement Series*, 268, 40. [DOI].

A. Mandal, D. Mukherjee, C. Federrath, N. P. H. Nesvadba, G. V. Bicknell, A. Y. Wagner, and M. Meenakshi (2021), "Impact of relativistic jets on the star formation rate: a turbulence-regulated framework," *Monthly Notices of the Royal Astronomical Society*, 508, 4738–4757. [DOI].

A. Mandal, C. Federrath, and B. Körtgen (2020), "Molecular cloud formation by compression of magnetized turbulent gas subjected to radiative cooling," *Monthly Notices of the Royal Astronomical Society*, 493, 3098–3113. [DOI].

A. Mandal and S. Nadkarni-Ghosh (2020), "One-point probability distribution function from spherical collapse: early dark energy versus ΛCDM," *Monthly Notices of the Royal Astronomical Society*, 498. [DOI].

Co-authored publications

S. Yeole, D. Mukherjee, and **A. Mandal** (2025), "Investigating field burial by magnetically confined accretion mounds on neutron stars," *Monthly Notices of the Royal Astronomical Society*, 541. [DOI].

M. Meenakshi, D. Mukherjee, A. Y. Wagner, N. P. H. Nesvadba, G. V. Bicknell, R. Morganti, R. M. J. Janssen, R. S. Sutherland, and **A. Mandal** (2022), "Modelling observable signatures of jet-ISM interaction: thermal emission and gas kinematics," *Monthly Notices of the Royal Astronomical Society*, 516. [DOI].

N. P. H. Nesvadba, A. Y. Wagner, D. Mukherjee, **A. Mandal**, R. M. J. Janssen, H. Zovaro, N. Neumayer, J. Bagchi, and G. Bicknell (2021), "Jet-driven AGN feedback on molecular gas and low star-formation efficiency in a massive local spiral galaxy with a bright X-ray halo," *Astronomy & Astrophysics*, 654. [DOI].

Manuscripts in preparation

A. Mandal et al., "What sets the thermal balance of fragmenting disks, and how robust is it to shock treatment?," in preparation.

A. Mandal et al., "A novel implementation of a subgrid model of star formation: Application to simulations of jet-ISM interaction," in preparation.

COMPETITIVE COMPUTING ALLOCATIONS

2026–2027 — **19.05 million CPU-hours**
The interplay of self-gravity, magnetic fields, and dust in protoplanetary disks
NHR Center at TU Dresden, Germany · PI

2023–2024 — **2.5 million CPU-hours**
Gadi, National Computational Infrastructure (NCI), Australia · Co-I

2022–2023 — **2.3 million CPU-hours**
Gadi, National Computational Infrastructure (NCI), Australia · Co-I

STUDENT SUPERVISION

Noa Marie Hoffmann *2024–Present*

Ph.D. student, Leibniz Institute for Astrophysics Potsdam (AIP), Germany
Co-supervisor; primary supervisor: Dr. Oliver Gressel.
Research: *Protostellar-outflow feedback and fragmentation in early-stage circumstellar/protoplanetary disks.*

Biplab Sarkar *2025–Present*

Ph.D. student, Inter-University Centre for Astronomy and Astrophysics, Pune, India
Co-supervisor; primary supervisor: Dr. Dipanjan Mukherjee.
Research: *Modelling the role of stellar and AGN feedback processes in galaxy evolution.*

EDUCATION

Ph.D. in Astrophysics *2019–2024*

Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India

Thesis: *Modelling the impact of AGN-driven outflows on the star formation activity in galaxies.*

Master's in Physics *2017–2019*

Indian Institute of Technology Kanpur, India

Thesis: *One-point probability distribution function from spherical collapse in different cosmologies.*

Bachelor's in Physics *2014–2017*

Ramakrishna Mission Residential College, Narendrapur, Kolkata, India

Gold Medal for ranking first in the graduating class.

AWARDS & FELLOWSHIPS

2019–2024 **Junior and Senior Research Fellowship**
Council of Scientific and Industrial Research (CSIR), India

May–Jul 2019 **Future Research Talent Travel Award**
Australian National University – collaborative research

2017 **Gold Medal**
Ramakrishna Mission Residential College, Narendrapur — overall topper in undergraduate study

2014–2019 **INSPIRE Fellowship**
Department of Science and Technology (DST), Government of India

SELECTED TALKS & PRESENTATIONS

July 2026 **"A fast spectral-Multigrid Poisson solver for self-gravity in global simulations of astrophysical disks"**
Dics on the Exe; University of Exeter, UK · Poster

Oct 2023 **"AGN Feedback: Negative or Positive – or Both?"**
Max Planck Institute for Astrophysics (MPA), Garching, Germany · Invited Seminar

Oct 2023 **"How do outflows from the AGN affect the star formation activity inside the host galaxy?"**
Lorentz Centre Workshop – The Importance of Jet-Induced Feedback on Galaxy Scales, Leiden, Netherlands · Invited Talk

Feb 2023 **"Effect of AGN-driven outflows on the star formation in galaxies"**
Galflows 2023, IUCAA, Pune, India · Contributed Talk

Sep 2022 **"Impact of relativistic jets on the star formation rate: a turbulence-regulated framework"**
31st Texas Symposium, Prague, Czech Republic · Contributed Talk

REFERENCES

Dr. Oliver Gressel
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Dr. Udo Ziegler
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